

# From Implicit Heuristics to Explicit Optimization: A Unified Framework for In-Context Relation Extraction

Xin Sun<sup>1,2\*</sup> (✉), Muyu Li<sup>1,2\*</sup>, Jianfei Zhao<sup>1,2</sup>, and Xinyang Liu<sup>1,2</sup>

<sup>1</sup> School of Computer Science and Technology, Beijing Institute of Technology, Beijing, China

<sup>2</sup> Beijing Engineering Applications Research Center on High Volume Language Information Processing and Cloud Computing, Beijing, China  
{sunxin, 3220241366, zhqingan, 3220220944}@bit.edu.cn

**Abstract.** In-context learning (ICL) for relation extraction is often undermined by its reliance on implicit heuristics. This weakness is critical in two stages: (1) example selection, where semantic similarity serves as a poor proxy for utility; (2) model reasoning, which depends on unguided attention mechanisms. This reliance on non-optimizable strategies leads to unreliable performance and hinders interpretability. This paper advocates for a fundamental shift from implicit heuristics to explicit optimization and guidance. We introduce **CLARE** (**C**ontribution-driven **L**earning and **A**ddaptive **R**Easoning), a unified framework that introduces core innovations to both the example selection and model reasoning stages. For example selection, we propose **Predictive Contribution Estimation (PCE)**, a novel method that trains a retriever to directly optimize for an example’s utility by quantifying its impact on the model’s prediction confidence. For model reasoning, we introduce **Adaptive Inference Modulation (AIM)**, which transforms the unguided inference process into a steerable one by dynamically modulating the model’s internal attention scores. This ensures that the most valuable demonstration information is precisely leveraged. Experiments on four challenging datasets validate that CLARE significantly outperforms mainstream ICL baselines by leveraging explicit contribution estimation and guided inference.

**Keywords:** Relation Extraction · In-context Learning · Contrastive Learning · Attention.

## 1 Introduction

In-Context Learning (ICL) has revolutionized few-shot Relation Extraction (RE) by enabling large language models (LLMs) to perform complex tasks without parameter updates [18, 1]. However, the success of this paradigm is built on a fragile foundation: a heavy reliance on implicit heuristics. This dependency is

---

\* These authors contributed equally to this work.

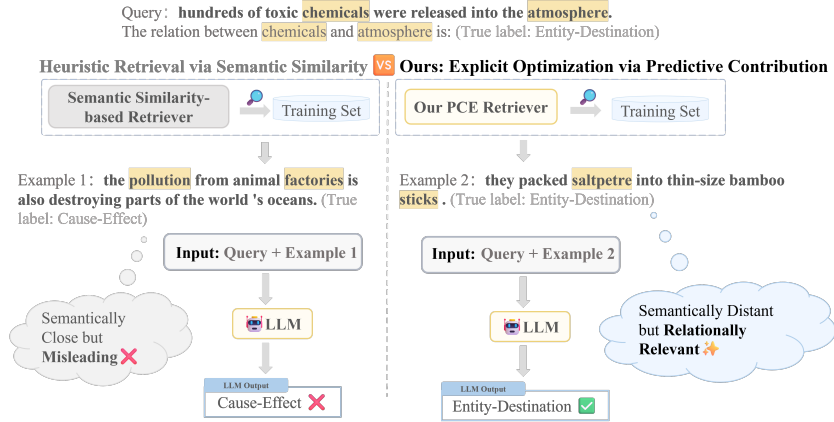


Fig. 1: Illustration of the core problem. **(left)** Conventional retrieval, relying on semantic similarity, selects a misleading example. Although topically similar—sharing terms like *pollution* and *factories* vs. *chemicals* and *atmosphere*—its relational pattern, *Effect-from-Cause*, fundamentally mismatches the query’s *Entity-released\_into-Destination* pattern, leading to an incorrect *Cause-Effect* prediction. **(right)** Our PCE approach, optimizing for contribution, identifies a semantically **distant** example, “packed saltpetre into... sticks”, that precisely shares the query’s core *Entity-<verb>-into-Destination* relational pattern, ensuring correct inference.

particularly detrimental in RE, where identifying relationships often hinges on subtle relational patterns rather than superficial semantic similarity. As illustrated in Figure 1, an example that is semantically close but demonstrates an irrelevant relational pattern can inject noise and mislead the model [8]. This leads to an ICL process for RE that is unguided and unstable at its two most critical stages: example selection and model inference.

Currently, many studies on example selection rely on semantic similarity (e.g., SimCSE [5], GPT-RE [18], C-ICL [13]), an indirect proxy for an example’s true utility, often failing as demonstrated in Figure 1 (left). Simultaneously, the inference process operates as a black box, relying on the LLM’s unguided attention mechanisms. This process is inherently unguided, making it difficult for the model to precisely leverage the most critical information **within** the provided examples, even if those examples are high-quality [2].

Motivated by the need to replace these implicit bottlenecks with transparent, objective-driven mechanisms, we introduce **CLARE** (Contribution-driven Learning and Adaptive REasoning), a unified framework that pioneers a shift from implicit heuristics to explicit optimization and guidance. First, to overcome unreliable example selection, CLARE introduces a *Predictive Contribution Estimation (PCE)* module. We quantify an example’s true value by its direct impact on prediction accuracy—its “predictive gain”—and use this explicit signal

to train a lightweight contrastive retriever, enabling the selection of truly relevant examples even if they appear semantically distant (Figure 1, right). Second, to address the black-box inference process, CLARE incorporates an *Adaptive Inference Modulation (AIM)* module. AIM replaces the model’s implicit, unguided reasoning with a transparent and steerable process that explicitly guides the model’s internal attention, ensuring that the most valuable information from the selected examples is effectively leveraged.

We conduct extensive experiments on four widely-used RE benchmarks (SemEval, SciERC, ACE05, and TACRED). The results validate that our explicit framework allows CLARE to consistently and significantly outperform mainstream ICL-based methods. Our main contributions are:

- We identify and articulate a core limitation of the current ICL paradigm for RE: its dependence on unguided, implicit heuristics for both example selection and model reasoning, illustrated by concrete failure cases (Figure 1).
- We propose Predictive Contribution Estimation (PCE), a novel method to train a retriever that explicitly optimizes for an example’s utility, moving beyond simple semantic similarity.
- We design Adaptive Inference Modulation (AIM), a mechanism that makes the LLM’s reasoning process controllable by explicitly guiding its internal attention.

## 2 Related Work

In-Context Learning (ICL) for relation extraction is powerful yet fragile, with its performance critically dependent on two stages still governed by implicit heuristics: example selection and model inference [8]. Our work challenges this paradigm by introducing explicit, optimization-driven mechanisms for both.

### 2.1 From Heuristic to Optimized Example Selection.

The dominant strategy for example selection relies on semantic similarity, exemplified by SimCSE [5], an unreliable proxy for an example’s true utility. Recognizing this limitation, more sophisticated heuristics have emerged. These include designing task-aware retrievers (GPT-RE [18]), employing dual-systems (DSARE [9]), or using complex feedback loops to verify reasoning and re-select examples (SRVF [7]). Other methods, like GL [14], augment the prompt with learned guidelines. Despite their ingenuity, these approaches remain fundamentally heuristic. Even more advanced paradigms that train a dedicated retriever [16,21] or leverage direct LLM feedback for preference learning (GenICL [23]) still rely on indirect, coarse-grained, or relative optimization signals (*e.g.*, ‘A is better than B’). These approaches fail to directly quantify an individual example’s intrinsic, absolute utility for a given query.

Our work proposes a different, non-preference-based approach. Our Predictive Contribution Estimation (PCE) module is designed to quantify an example’s

absolute, quantifiable utility (its predictive gain). This allows us to train a retriever using a supervised, contrastive framework that explicitly distinguishes between demonstrably helpful and harmful examples, rather than just learning their relative order.

## 2.2 From Black-Box to Steerable Inference.

The second critical weakness of ICL is treating the LLM’s inference process as a black box. The standard approach of simply concatenating examples relies on the model’s unguided attention mechanisms to distill relevant information [18,13]. This is often suboptimal, as noisy demonstrations can mislead the model, sometimes resulting in performance worse than a zero-shot baseline [15]. While frameworks like SRVF [7] attempt to verify the model’s reasoning post-hoc, they do not make the internal inference process itself transparent or steerable. A key research gap remains in developing methods that can actively guide the model’s focus on salient information within examples [19].

Our Adaptive Inference Modulation (AIM) module directly addresses this gap. It transforms the unguided inference process into a transparent and controllable one by explicitly modulating the model’s internal attention scores, ensuring that the most valuable demonstration information is precisely leveraged.

## 3 Methodology

In-context learning (ICL) for relation extraction is fundamentally constrained by its reliance on two implicit and unguided processes: example selection and inference. To address these limitations, we propose **CLARE**, a unified framework designed to systematically replace these implicit heuristics with explicit, optimizable modules. As illustrated in Figure 2, our framework consists of two key stages: (1) Explicit Modeling of Example Contribution for retrieval, and (2) Explicit Guidance of the Inference Process.

### 3.1 Task Formulation

We first formulate the standard ICL-based RE task. Given a query  $x_q$  and a large candidate pool of available examples  $\mathcal{C}$ , the initial goal is to select an ordered set of  $N$  demonstrations  $E = \{e_1, \dots, e_N\}$  where  $e_i \in \mathcal{C}$ . These demonstrations are then formatted and concatenated with the query to form a single prompt,  $\mathcal{P} = (e_1, \dots, e_N, x_q)$ . This prompt is fed to a pre-trained LLM, which autoregressively generates the relation  $y$ . The objective is to maximize the conditional probability of the true relation  $y_{true}$ :

$$\operatorname{argmax} P(y_{true} | e_1, \dots, e_N, x_q) \quad (1)$$

The overall performance critically depends on two independent sub-problems: (1) the **selection** mechanism for choosing the optimal set  $E$ , and (2) the **inference** process by which the model utilizes  $E$  to compute the probability. Our work introduces explicit optimization for both.

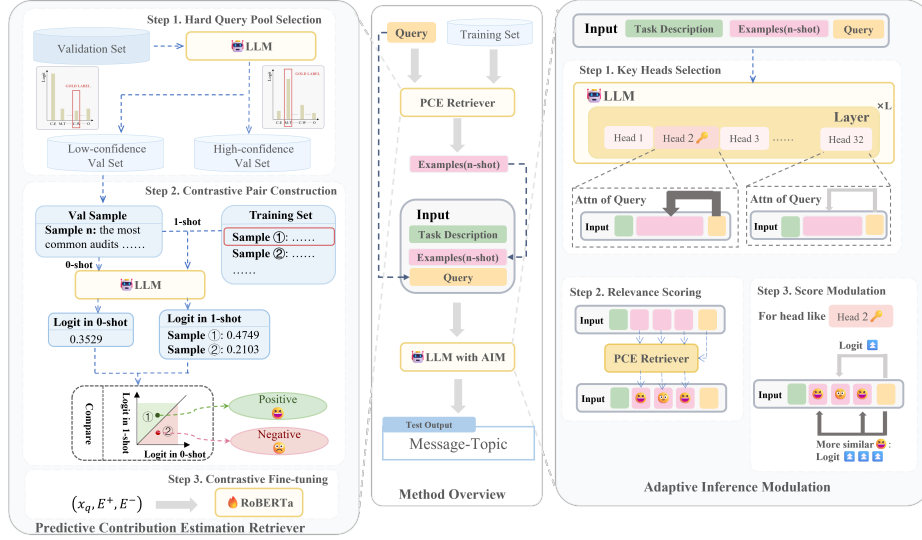


Fig. 2: The overall architecture of our proposed CLARE framework.

### 3.2 Predictive Contribution Estimation (PCE)

The effectiveness of ICL hinges on example quality, but this process is typically governed by the implicit heuristic of semantic similarity—an unreliable proxy for an example’s true utility. To replace this heuristic with a principled optimization target, we introduce **Predictive Contribution Estimation (PCE)**, a framework for training a retriever on an explicit objective. The core idea is to directly measure an example’s value by its **quantifiable impact** on the model’s predictive performance, as detailed in Figure 2 (left).

**Quantifying Example Contribution.** Conventional heuristics like **relation label consistency** are flawed, as they fail to capture an example’s true utility. To overcome this, we propose quantifying contribution based on its direct impact on the LLM’s predictive behavior. We introduce the **Predictive Contribution Score (PCS)**, an explicit metric measuring the change in the model’s confidence in the correct label  $y_{true}$  when presented with a candidate example  $e_c$ .

Formally, for a given query  $x_q$ , its PCS is defined as:

$$PCS(e_c, x_q) = P(y_{true}|e_c, x_q) - P(y_{true}|x_q) \quad (2)$$

where:

- $P(y_{true}|x_q)$  is the baseline probability (zero-shot).
- $P(y_{true}|e_c, x_q)$  is the probability (one-shot) with  $e_c$  as context.

In cases where a label  $y_{true}$  consists of multiple tokens (e.g., "Cause-Effect"),  $P(y_{true}|\dots)$  is defined as the length-normalized average log-probability of the token sequence  $y = (y_1, \dots, y_T)$ :

$$\log P(y_{true}|\dots) = \frac{1}{T} \sum_{t=1}^T \log P(y_t|y_{<t}, \dots) \quad (3)$$

This normalization ensures that the probabilities of labels with different lengths are comparable.

This score provides an explicit measure of utility: a positive PCS identifies a **positive contributor**, while a negative PCS identifies a **negative contributor**. While PCS is defined in 1-shot to isolate individual contribution, we find this utility signal generalizes effectively to n-shot scenarios (see Experiments).

**Contrastive Pair Construction.** Armed with the PCS metric, we devise an automated pipeline to construct a training dataset  $(x_q, E^+, E^-)$ . As shown in Figure 2 step 1 & 2, we first identify low confidence queries  $x_q$  from the development set, defined as those where the 0-shot prediction is either incorrect or correct but with low confidence ( $P(y_{true}|x_q) < \delta_{conf}$ ). For each  $x_q$ , we pair it with a candidate pool  $e_c$  sampled from the training set. To ensure computational scalability, we do not iterate over the entire corpus; instead, we construct a representative subset  $\mathcal{C}' \subset \mathcal{C}$  for each query (details in Sec. 4.1).

A threshold  $\delta > 0$  is then applied to create the explicit **Positive Set** ( $E^+$ ), defined by examples where  $PCS > \delta$ , and the **Negative Set** ( $E^-$ ), defined by examples where  $PCS < -\delta$ . This process discards negligible examples.

**Training.** We train a contribution-aware retriever,  $f_\theta$ , using the constructed pairs. Our approach employs a supervised contrastive objective, InfoNCE loss. This is superior to standard methods because our negative samples  $E^-$  are explicitly identified as harmful, providing a stronger, task-relevant signal than random in-batch negatives.

$$\mathcal{L}_{cl} = -\log \frac{\sum_{i=1}^N \exp(\text{sim}(\mathbf{h}_q, \mathbf{h}_i^+)/\tau)}{\sum_{i=1}^N \exp(\text{sim}(\mathbf{h}_q, \mathbf{h}_i^+)/\tau) + \sum_{j=1}^M \exp(\text{sim}(\mathbf{h}_q, \mathbf{h}_j^-)/\tau)} \quad (4)$$

Upon training, the PCE retriever is equipped to select the top-k most beneficial examples for the next stage: inference.

### 3.3 Adaptive Inference Modulation (AIM)

PCE provides high-quality examples, but the LLM’s reasoning remains an unguided black box. To fix this, we introduce the **Adaptive Inference Modulation (AIM)** module, which transforms inference into a guided procedure. AIM’s core principle is to dynamically modulate self-attention scores. This modulation is not uniform; it is intelligently weighted using the relevance scores from the

same PCE retriever, creating a synergistic framework where the explicit utility signal is used for both selection and inference. The process unfolds in three key steps:

**Key Attention Head Selection.** Given the known **functional specialization** of attention heads [4,20], modifying all of them uniformly is suboptimal. AIM therefore performs a targeted intervention. To identify the most critical heads, we first quantify each head’s contribution to the information flow between the query (token indices  $T_q$ ) and the context (token indices  $T_c$ ) within a target layer  $l$ . We calculate its **Query-Context Interaction Score**,  $S^{(l)}(h)$ , using the post-softmax attention probabilities,  $\bar{A}_h^{(l)}$ :

$$S^{(l)}(h) = \frac{1}{|T_q| \cdot |T_c|} \sum_{j \in T_q} \sum_{k \in T_c} \bar{A}_{h,j \rightarrow k}^{(l)} \quad (5)$$

where  $\bar{A}_{h,j \rightarrow k}^{(l)}$  is the normalized probability computed from the pre-softmax scores  $A_h^{(l)}$ :

$$\bar{A}_{h,j \rightarrow k}^{(l)} = \text{Softmax}_{k'} \left( A_{h,j \rightarrow k'}^{(l)} \right)_k \quad (6)$$

where  $A_{h,j \rightarrow k}^{(l)}$  is the attention score from token  $j$  to token  $k$ , in head  $h$  of layer  $l$ .

We then select the set  $\mathcal{H}_{qc}^{(l)}$ , which comprises the top- $k$  percent of heads based on this score. This ensures that our modulation is applied only to the heads most active in integrating query-context information.

**Relevance-Weighted Guidance Scoring.** Next, to determine the magnitude of modulation for each example, we quantify the relevance of each selected example  $e_i$  to the query  $x_q$ . We directly leverage the PCE retriever  $f_\theta$  for this task, which establishes the synergy between our selection and inference modules:

$$s_i = \text{sim}(f_\theta(x_q), f_\theta(e_i)) \quad (7)$$

These raw similarity scores, collected into a vector  $\mathbf{s} = (s_1, \dots, s_N)$ , are then normalized via softmax to produce the final guidance weight vector  $\mathbf{w} = (w_1, \dots, w_N)$ :

$$\mathbf{w} = \text{Softmax}(\mathbf{s}) \quad (8)$$

**Explicit Attention Score Modulation.** Finally, with the target heads  $\mathcal{H}_{qc}^{(l)}$  identified and the guidance weights  $\mathcal{W}$  computed, we apply the modulation. We compute a specific modulation value  $\Delta A_i$  for each example  $e_i$ . This value incorporates its guidance weight  $w_i$ , a positional decay term  $\left(\frac{N-i}{N}\right)$  that prioritizes earlier examples, and a strength hyperparameter  $\alpha_{AIM}$ :

$$\Delta A_i = \alpha_{AIM} \cdot w_i \cdot \left( \frac{N-i}{N} \right) \quad (9)$$

This modulation value is then directly added to the pre-softmax attention scores of the targeted heads  $\mathcal{H}_{qc}^{(l)}$ . Specifically, for any head  $h \in \mathcal{H}_{qc}^{(l)}$ , the score from any token  $t_j$  to any token  $t_k$  within the span of example  $e_i$  (denoted  $T_{e_i}$ ) is updated as:

$$A'_{h,j \rightarrow k}^{(l)} = A_{h,j \rightarrow k}^{(l)} + \Delta A_i, \quad \forall k \in T_{e_i} \text{ and } \forall j \text{ where } j > k \quad (10)$$

This rule forces the model to "look back" more strongly at the demonstrations PCE identified as most valuable, completing our shift to an explicit guidance pipeline.

## 4 Experiment

### 4.1 Experimental Setup

**Datasets.** We conduct experiments on four widely-recognized relation extraction benchmarks: SemEval-2010 Task 8 [6], SciERC [12], ACE05 [17], and TACRED [22]. We use Micro-F1 as the primary evaluation metric across all datasets.

- **SemEval-2010 Task 8** is a seminal benchmark for sentence-level relation extraction, containing 9 directed semantic relations and an "Other" category.
- **SciERC** is sourced from scientific abstracts, featuring 7 relation types and presenting the challenge of cross-sentence relations.
- **ACE05** is a complex dataset derived from diverse sources, defining 7 coarse-grained entity relation types.
- **TACRED** is a large-scale dataset with 41 fine-grained relation types from the political domain. Since over 80% of instances are negative ("no\_relation"), we follow standard protocols and report Micro-F1 scores exclusively on positive relation types.

The statistics for each dataset are summarized in Table 1.

Table 1: Statistics of the datasets used in our experiments. The '+1' in SemEval-2010 denotes the "Other" relation type.

| Dataset      | Train  | Dev    | Test   | Total   | #Relations |
|--------------|--------|--------|--------|---------|------------|
| SemEval-2010 | 6,507  | 1,493  | 2,717  | 10,717  | 9+1        |
| SciERC       | 1,861  | 275    | 551    | 2,687   | 7          |
| ACE05        | 10,051 | 2,050  | 2,420  | 14,521  | 7          |
| TACRED       | 68,124 | 15,509 | 22,631 | 106,264 | 41         |

**Baselines.** We select the following models as our baselines. For fairness and due to experimental costs, we reproduced the models that originally used GPT with Llama-3-8B-Instruct, and all experiments for both reproduced baselines and our method were conducted under a unified 10-shot setting.



- **Random** selects in-context demonstrations from the labeled samples, serving as a foundational baseline without any specific retrieval strategy.
- **GL** [14] proposes Guideline Learning (GL), which learns from errors to create guidelines that are retrieved at inference time to mitigate underspecified task descriptions.
- **SimCSE** [5] is a strong baseline that uses a simple contrastive learning framework to learn robust sentence embeddings.
- **GPT-RE** [18] enhances in-context learning through task-aware retrieval and gold label-induced reasoning.
- **SRVF** [7] trains a rationale supervisor to verify the LLM’s reasoning and provides re-selected demonstrations as feedback to correct biased predictions.
- **DSARE** [9] introduces a dual-system approach to mutually enhance a traditional RE model and an LLM, where the LLM augments training data and the RE model retrieves valuable demonstrations for in-context learning.

**Implementation Details.** We employ Meta-Llama-3-8B-Instruct [3] as our backbone model for the final relation prediction. For the retriever, we initialize its encoder with RoBERTa-large [10]. All experiments are conducted on a server equipped with eight NVIDIA RTX 3090 Ti GPUs. The retriever is trained for 400 epochs using the AdamW [11] optimizer with a learning rate of 1e-6. We use a batch size of 1 and 4 gradient accumulation steps.

To build the PCE training set, we identified low-confidence queries and computed their PCS scores against a candidate pool. To address the computational cost of this initial phase, we applied a downsampling strategy: for each query, we computed scores against 2,000 randomly sampled candidates rather than the full dataset. We opted for random sampling over coarse semantic filtering to ensure the retriever learns from a diverse range of utility distributions, preventing bias towards semantic similarity. Although building this set requires LLM forward passes, it is a one-time offline cost that enables a lightweight, highly efficient retriever for all future inference, ensuring practical scalability. The threshold  $\delta$  for constructing positive and negative samples is set to 0.01, and the temperature parameter  $\tau$  for contrastive learning is 0.05.

To constrain the output format and improve prediction reliability, we append the template *The correct answer is: \*\** to the model’s input. The model’s prediction is then determined by the logits of the token immediately following this template. To compute label probabilities for PCS, we apply a softmax function over the logits of only the valid relation label tokens for that specific task.

During inference, we generate 3 randomly shuffled candidate example orders for each sample, and the final prediction is derived from the mean of these logits across the 3 orders. For a fair comparison, this 3-shuffle ensembling strategy was applied to all methods. For CLARE, the AIM module, including its positional decay, was re-computed independently for each of the 3 permutations.

Table 2: Main results.

| Methods             | SemEval      | TACRED       | SciERC       | ACE05        | Avg.         |
|---------------------|--------------|--------------|--------------|--------------|--------------|
| Random              | 49.91        | 59.58        | 60.16        | 38.58        | 52.06        |
| GL [14]             | <b>65.90</b> | 63.60        | 65.55        | 52.28        | <u>61.83</u> |
| SimCSE [5]          | 58.30        | 62.74        | 63.35        | <b>54.13</b> | 59.63        |
| GPT-RE [18]         | 64.93        | <u>63.45</u> | 65.57        | 51.92        | 61.47        |
| SRVF [7]            | 64.13        | 62.58        | 65.49        | 52.02        | 61.06        |
| DSARE [9]           | 65.25        | 63.38        | <u>65.62</u> | 52.20        | 61.61        |
| <b>CLARE (ours)</b> | <u>65.62</u> | <b>64.02</b> | <b>65.71</b> | <u>52.30</u> | <b>61.91</b> |

## 4.2 Main Results

Table 2 presents the main experimental results on four benchmark datasets. Our proposed framework, CLARE, consistently demonstrates superior performance, achieving the highest average Micro-F1 score of 61.91 across all benchmarks. This result validates the effectiveness of our explicit, optimization-driven approach for in-context relation extraction.

Compared to the baselines, CLARE shows significant improvements. It surpasses the Random selection strategy by a large margin (an average of +9.85 F1 points), underscoring the necessity of a principled example selection method. More importantly, CLARE outperforms SimCSE, a strong baseline relying on semantic similarity, on three out of four datasets. This highlights the core limitation of using similarity as a proxy for utility and validates our hypothesis that directly optimizing for predictive contribution yields more effective demonstrations. Notably, SimCSE is the top performer on ACE05, outperforming all other methods including ours. We hypothesize this is due to the unique characteristics of the dataset itself. ACE05 defines only 7 coarse-grained relation types (*e.g.*, *PHYS* for physical, *PER-SOC* for personal-social), where high semantic or topical similarity—which SimCSE excels at capturing—is often a reliable proxy for an example’s true utility.

This scenario, where the proxy heuristic (similarity) aligns well with the true objective (utility), may explain why SimCSE excels. However, CLARE’s superior and consistent performance on the other three datasets, especially its state-of-the-art results on the fine-grained TACRED (64.02) and SciERC (65.71), validates its robustness. It demonstrates that CLARE excels precisely on tasks where the semantic heuristic fails and a more explicit, contribution-driven model is necessary.

## 4.3 Ablation Study

To isolate the contributions of our core components, we conducted a comprehensive ablation study, with results presented in Table 3. The analysis reveals that the Predictive Contribution Estimation (PCE) retriever is the primary driver of performance. Replacing the standard SimCSE retriever with our PCE module

Table 3: Ablation study of CLARE’s components. **CLARE (Full Model)** combines our **PCE Retriever** with **AIM Inference**. The ablations replace one component with a baseline: **CLARE w/o AIM** uses our **PCE Retriever** with **Standard Inference**; **CLARE w/o PCE** uses **Random Selection** with our **AIM Inference**. Baselines include **Standard ICL (SimCSE Retriever + Standard Inference)** and **Random Selection (Random Selection + Standard Inference)**.

| Configuration             | SemEval      | TACRED       | SciERC       | ACE05        |
|---------------------------|--------------|--------------|--------------|--------------|
| <b>CLARE (Full Model)</b> | <b>65.62</b> | <b>64.02</b> | <b>65.71</b> | <b>52.30</b> |
| CLARE w/o AIM             | 65.37        | 63.66        | 65.40        | 52.13        |
| CLARE w/o PCE             | 50.61        | 59.44        | 60.37        | 39.01        |
| Standard ICL              | 58.30        | 62.74        | 63.35        | 54.13        |
| Random Selection          | 49.91        | 59.58        | 60.16        | 38.58        |

(CLARE w/o AIM) yields a substantial F1 score improvement across all datasets (*e.g.*, from 58.30 to 65.37 on SemEval), powerfully validating that directly optimizing for predictive utility is superior to the heuristic of semantic similarity. While PCE provides the foundational boost, the Adaptive Inference Modulation (AIM) module offers further, consistent gains by refining the reasoning process. Adding AIM to the PCE retriever (CLARE (Full Model)) consistently elevates performance, confirming that explicitly guiding the model’s attention helps it leverage the high-quality demonstrations more effectively. Crucially, the synergy between the two is vital: AIM offers minimal benefit when paired with random selection (CLARE w/o PCE), proving that an advanced inference mechanism cannot compensate for low-quality inputs. This underscores the success of our two-stage explicit optimization strategy, where PCE first identifies valuable examples and AIM then ensures their influence is maximized.

#### 4.4 Analysis on AIM and PCE

To validate the explicit optimization strategies of our two core modules, we conducted detailed component analyses for both PCE and AIM.

*Analysis of PCE (Case Study).* To provide a qualitative understanding of PCE’s superiority over semantic heuristics, we analyze its selections for a challenging query, as shown in Table 4. The query requires an *Entity-Destination* relation for *...stored the **seeds** in a... **unit***. A semantic retriever like SimCSE would fail here, as it would search for topically-related "planting" examples and miss the high-utility candidates shown, both of which have very low semantic similarity (0.185 and 0.177).

Our PCE module, optimizing for predictive contribution, demonstrates two key advantages. First, as seen in **Example 1**, it successfully discovers a semantically distant example (*...residents... moved **into**... **home***) that SimCSE would

Table 4: Case Study: PCE vs. Semantic Similarity (SimCSE). We analyze two examples selected by PCE for a single query, both of which SimCSE would miss. The head and tail entities are bolded.

| Example                                                                                                                                  | PCE & Sim                              |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <i>Query (GT: Entity-Destination)</i>                                                                                                    |                                        |
| Fall planting can not be completed before the ground freezes, so I stored the <b>seeds</b> in a <b>stratification unit</b> until spring. |                                        |
| <i>Example 1 (Label: Entity-Destination)</i>                                                                                             |                                        |
| The new <b>residents</b> have been moved into their brand new <b>home</b> .                                                              | <b>PCE:</b> 0.632<br><b>Sim:</b> 0.185 |
| <i>Example 2 (Label: Origin-Entity)</i>                                                                                                  |                                        |
| The program featured <b>clips</b> from 1950's <b>television</b> and interviews with game show stars.                                     | <b>PCE:</b> 0.648<br><b>Sim:</b> 0.177 |

Table 5: Ablation study of the Adaptive Inference Modulation (AIM) module on the SemEval dataset. We ablated each of the three components in turn: the Weighted Guidance, the Targeted Heads modulation, and the Positional Decay.

| AIM Configuration                              | F1 Score     |
|------------------------------------------------|--------------|
| <b>Full AIM (in CLARE)</b>                     | <b>65.62</b> |
| <i>Ablation of AIM Components</i>              |              |
| w/o Weighted Guidance<br>(Uniform Modulation)  | 65.44        |
| w/o Targeted Heads<br>(Untargeted Modulation)  | 65.39        |
| w/o Positional Decay<br>(No Positional Factor) | 65.44        |

have missed. PCE identifies it as valuable (Score: 0.63) because it shares the identical *Entity-Destination* label and a similar relational structure.

More powerfully, **Example 2** validates our hypothesis that predictive utility is superior to simple label consistency. This example (...*clips from... television...*) has a completely *different* relation label (*Origin-Entity*). Yet, PCE assigns it an even *higher* contribution score (0.65). This indicates that this example’s relational pattern, despite its label mismatch, provides a more potent signal for boosting the model’s confidence on the query than Example 1. This case strongly demonstrates that PCE’s explicit optimization allows it to uncover a more diverse and effective set of examples, including those that are semantically distant and even those with different labels, which traditional methods would erroneously discard.

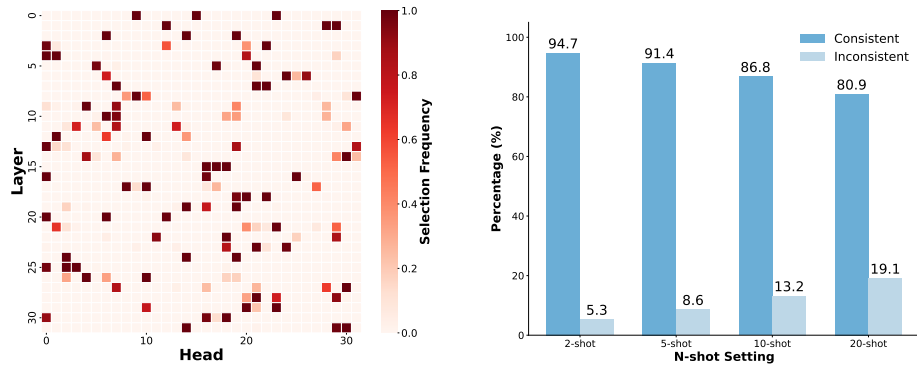
*Analysis of AIM.* Table 5 presents a component-wise ablation of the AIM module on the SemEval dataset. The full CLARE model, equipped with the complete AIM, achieves an F1 score of 65.62. Removing any single component results in a performance drop, validating their contribution. The most significant drop (to 65.39) occurs when we remove “Targeted Heads” and apply modulation uniformly. This result strongly supports our hypothesis of functional specialization, confirming that strategically intervening on the specific heads responsible for query-context interaction is far more effective than a naive, untargeted boost. Removing the “Weighted Guidance” (using uniform modulation instead of PCE

scores) or the "Positional Decay" (treating all positions equally) also degrades performance to 65.44. This shows that both leveraging the retriever’s relevance scores and prioritizing examples earlier in the context are valuable strategies. In summary, all three components of AIM are integral to its success.

#### 4.5 Analysis of Positive Example Composition

To validate our 1-shot-based sample construction, we evaluated whether the positive effect of examples generalizes when they are composed in n-shot contexts. We conducted an experiment on the SemEval dataset, measuring the consistency of prompts composed of 2, 5, 10, and 20 examples previously identified as positive. Consistency is defined as the combined prompt still improving the correct label’s logit score.

The results demonstrate remarkable robustness. The consistency rate declines gracefully from **94.7%** (2-shot) to a strong **80.9%** even in a dense 20-shot scenario. This gradual and expected decrease, caused by rising inter-example noise in larger contexts, confirms that our 1-shot identification strategy is fundamentally sound. It successfully provides a reliable data foundation for training a retriever that identifies examples robustly beneficial in realistic, multi-example settings.



(a) Frequency of key attention head selection on the SemEval test set.

(b) Consistency Statistics of Reasoning Results under different n-shot settings.

Fig. 3: Comprehensive analysis of model behavior under In-Context Learning (ICL). (a) Frequency of key attention head selection on the SemEval test set. (b) Reasoning Consistency Statistics detail the percentage of consistent and inconsistent results across various shot settings.

#### 4.6 Empirical Validation of Targeted Head Identification

To validate our data-driven approach for identifying critical attention heads, we analyzed their selection frequency on the SemEval test set using **Llama3-8B-Instruct**. We designated the top 10% of heads in each layer as critical based on their Query-Context Interaction Scores and aggregated the counts across all test cases.

The resulting heatmap (Figure 3a) reveals two crucial patterns. First, within any given layer, selection is highly concentrated on a small, consistent subset of heads, confirming functional specialization. Second, and more critically, this set of important heads is dynamic and changes across different layers. This strongly indicates a hierarchical specialization, where the model leverages different heads for reasoning at different processing depths.

This finding powerfully justifies AIM’s core design: a targeted, layer-specific intervention is not merely an optimization but a necessity to adapt to the evolving functional roles of attention heads throughout the network.

### 5 Conclusion

In-context learning (ICL) for relation extraction is often undermined by its reliance on implicit heuristics. This paper identified two critical bottlenecks—heuristic example selection based on flawed proxies like semantic similarity, and unguided model reasoning. We argued for a paradigm shift from these implicit heuristics to explicit modeling and guidance.

To this end, we introduced **CLARE**, a unified framework with two synergistic components. First, Predictive Contribution Estimation (PCE) trains a retriever to directly optimize for an example’s true utility by quantifying its predictive gain. Second, Adaptive Inference Modulation (AIM) transforms the black-box inference process into a steerable one by dynamically modulating the model’s internal attention, ensuring that the high-quality examples selected by PCE are precisely leveraged.

Extensive experiments on four datasets demonstrated that our explicit, optimization and guidance approach allows CLARE to consistently outperform strong ICL baselines. This work validates the effectiveness of replacing fragile heuristics with principled, optimizable mechanisms and opens new avenues for building more robust, controllable, and interpretable in-context learning systems.

**Acknowledgments.** This paper is supported by the Monitoring and Early Warning Service Platform Project (2023-275-1-1).

### References

1. Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Nee-lakantan, A., Shyam, P., Sastry, G., Askell, A., et al.: Language models are few-shot learners. In: Advances in Neural Information Processing Systems. vol. 33, pp. 1877–1901 (2020)

2. Dong, Q., Li, L., Dai, D., Zheng, C., Ma, Z., Li, R., Xia, H., Xu, J., Wu, Z., Liu, B., et al.: A survey on in-context learning. arXiv preprint arXiv:2301.00234 (2023)
3. Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Yang, A., Fan, A., et al.: The Llama 3 herd of models. arXiv preprint arXiv:2407.21783 (2024)
4. Elhelo, A., Geva, M.: Inferring functionality of attention heads from their parameters. arXiv preprint arXiv:2412.11965 (2024)
5. Gao, T., Yao, X., Chen, D.: SimCSE: Simple contrastive learning of sentence embeddings. In: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing. pp. 6894–6910 (2021)
6. Hendrickx, I., Kim, S.N., Kozareva, Z., Nakov, P., Séaghdha, D.O., Padó, S., Pennacchiotti, M., Romano, L., Szpakowicz, S.: SemEval-2010 Task 8: Multi-way classification of semantic relations between pairs of nominals. In: Proceedings of the 5th International Workshop on Semantic Evaluation. pp. 33–38 (2010)
7. Li, Y., Miao, X., Zhou, S., Xu, M., Ren, Y., Qian, T.: Enhancing relation extraction via supervised rationale verification and feedback. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 39, pp. 24521–24529 (2025)
8. Liu, J., Shen, D., Zhang, Y., Dolan, B., Carin, L., Chen, W.: What makes good in-context examples for GPT-3? In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 100–114 (2022)
9. Liu, Y., Zhang, K., Gan, A., Yue, L., Hu, F., Liu, Q., Chen, E.: Empowering few-shot relation extraction with the integration of traditional RE methods and large language models. In: Database Systems for Advanced Applications. Lecture Notes in Computer Science, vol. 14600, pp. 349–359. Springer (2024)
10. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: RoBERTa: A robustly optimized BERT pretraining approach. arXiv preprint arXiv:1907.11692 (2019)
11. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: International Conference on Learning Representations (2019)
12. Luan, Y., He, L., Ostendorf, M., Hajishirzi, H.: Multi-task identification of entities, relations, and coreference for scientific knowledge graph construction. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. pp. 3219–3232 (2018)
13. Mo, Y., Liu, J., Yang, J., Wang, Q., Zhang, S., Wang, J., Li, Z.: C-ICL: Contrastive in-context learning for information extraction. arXiv preprint arXiv:2402.11254 (2024)
14. Pang, C., Cao, Y., Ding, Q., Luo, P.: Guideline learning for in-context information extraction. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. pp. 14435–14452 (2023)
15. Peng, K., Ding, L., Yuan, Y., Liu, X., Zhang, M., Ouyang, Y., Tao, D.: Revisiting demonstration selection strategies in in-context learning. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 12850–12869 (2024)
16. Rubin, O., Herzig, J., Berant, J.: Learning to retrieve prompts for in-context learning. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 2655–2671 (2022)
17. Walker, C., Strassel, S., Medero, J., Maeda, K.: ACE 2005 multilingual training corpus, linguistic Data Consortium, Philadelphia (2006)

18. Wan, Z., Cheng, F., Mao, Z., Liu, Q., Song, H., Li, J., Kurohashi, S.: GPT-RE: In-context learning for relation extraction using large language models. In: Proceedings of the 31st ACM International Conference on Multimedia. pp. 7634–7643 (2023)
19. Wang, L., Li, L., Dai, D., Chen, D., Zhou, H., Meng, F., Zhou, J., Sun, X.: Label words are anchors: An information flow perspective for understanding in-context learning. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. pp. 9840–9855 (2023)
20. Yang, H., Cho, H., Inoue, N.: Localizing task recognition and task learning in in-context learning via attention head analysis. arXiv preprint arXiv:2509.24164 (2025)
21. Ye, J., Wu, Z., Feng, J., Yu, T., Kong, L.: Compositional exemplars for in-context learning. In: International Conference on Machine Learning. pp. 39818–39833. PMLR (2023)
22. Zhang, Y., Zhong, V., Chen, D., Angeli, G., Manning, C.D.: Position-aware attention and supervised data improve slot filling. In: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing. pp. 35–45 (2017)
23. Zhang, Z., Lan, S., Song, L., Bian, J., Li, Y., Ren, K.: Learning to select in-context demonstrations preferred by large language model. arXiv preprint arXiv:2505.19966 (2025)